Not Too Hot to Run: Adaptation to Extreme Heat in Competitive Runners *

Fernando M. Aragón Mahyar Rezazadeh

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Abstract

This paper examines how adaptation shapes the impact of extreme heat on labor productivity. We use global data on competitive runners, combined with high-resolution weather information, to estimate how race-day heat affects running speed. Consistent with adaptation, we find that runners from hotter countries are significantly less affected by extreme temperatures. The performance gap is most pronounced in endurance events and narrows in high-stakes races, suggesting that both physiological factors and incentives may play a role. Our findings contribute to the literature on climate adaptation by providing evidence that prior exposure reduces sensitivity to heat.

JEL classification: Q54, J24.

Keywords: adaptation, heat stress, labor productivity, extreme heat, human performance.

* *Contact Information:* Aragón: Department of Economics, Simon Fraser University, Burnaby, BC, V5A 1S6, Canada (e-mail: faragons@sfu.ca); Rezazadeh: Department of Economics, Simon Fraser University; email: smr20@sfu.ca

1 Introduction

A growing body of evidence suggests that human adaptation can partially mitigate the harmful socio-economic effects of high temperatures. Several studies, for example, show that the relationship between temperature and mortality varies with income levels, baseline climate conditions, and access to cooling technologies such as air conditioning (Carleton et al., 2022; Barreca et al., 2016).

However, much less is known about the role of adaptation in mitigating the impact of extreme heat on labor productivity. This is an important limitation given the expected increase in global temperatures and the existing medical and economic evidence on the negative effect of heat on human physiology, performance, and labor productivity, especially in occupations that involve outdoor physical activities (Zhao et al., 2021; Sexton et al., 2022; Schlenker and Roberts, 2009).

Evidence on acclimatization, primarily drawn from small-scale controlled experiments, suggests that repeated short-term exposure can reduce the sensitivity of human physiological responses to extreme temperatures. Although informative, this evidence may not fully capture the effects of long-term adaptation to heat and may overlook the role of incentives, which could influence performance in real-world settings. Recent economic studies have begun to explore long-term adaptation to heat and human performance using observational data from athletes (e.g., tennis players, archers), but the evidence remains inconclusive (Burke et al., 2023; Qiu and Zhao, 2022).

This paper contributes to the debate by using a new global dataset of competitive runners to examine how adaptation shapes the effects of heat on human performance. To conduct our empirical analysis, we compile data from running events (100m to 10,000m races) sanctioned by the World Athletics Organization (WAO) and held worldwide between 2005 and 2023. We combine these records with reanalysis data to construct a panel dataset that includes

information on runners' speed (our measure of performance) and local environmental conditions on race day. This comprehensive dataset allows us to estimate the relationship between running speed and temperature, exploiting within-runner variation while controlling for a rich set of environmental factors. To examine adaptation, we allow the temperature-speed relationship to vary with the climate of a runner's country of nationality.

We begin by documenting the relationship between running speed and temperature. Consistent with previous research, we find an inverted U-shaped pattern: performance peaks around 30 ℃and declines at higher temperatures. However, we find that runners from warmer countries experience significantly smaller declines in speed in extreme heat than those from colder countries. We interpret this reduced sensitivity as evidence of long-term adaptation to heat.

Adaptation appears to play a quantitatively important role. Our estimates suggest that increasing air temperature from 30 ℃ to 40 ℃ would reduce the speed of a runner from a cold country by nearly 0.10 standard deviations, while having no effect on runners from warmer climates. This difference in speed is comparable to the typical gap between gold and silver medalists.

We find that sensitivity to extreme heat and the corresponding gains from adaptation are more pronounced in medium- and long-distance races. These results align with previous evidence suggesting that heat primarily affects aerobic activities based on endurance (Gibson et al., 2020). We also show that heat sensitivity decreases in higher-stakes races, such as finals, for runners from both hot and cold countries. We interpret this pattern as suggestive evidence that both physiological factors and incentives may influence performance. In particular, it is consistent with runners exerting greater effort in finals, which partially offsets the negative effects of heat.[1]

[1]There is evidence that individuals can improve their responses to environmental stressors, such as heat and air pollution, through increased psychological resilience, even in the absence of physiological adaptations (Evans et al., 1982; Blanchfield et al., 2014).

Finally, we find no evidence that heat sensitivity varies by runners' characteristics such as elite status (measured by relative overall performance) or sex. Our results also remain robust when controlling for the GDP of the runner's country of origin and the altitude of the competition venue. These findings reduce concerns that observed differences in heat sensitivity, interpreted as evidence of adaptation, are driven by other sources of heterogeneity.

This paper contributes to a growing economic literature that examines how short- and long-term adaptation influences the impact of heat on human performance. Similarly to our approach, several recent studies use observational data and focus on athletes, but the findings remain mixed. For example, Burke et al. (2023) show that extreme heat impairs tennis player performance but finds no differences in heat sensitivity between players born or residing in warmer versus cooler climates. Qiu and Zhao (2022), using data on archers in China, also find no relationship between heat sensitivity and the climate of the athlete's city of residence. However, they do observe differences based on birthplace, comparing athletes born in northern versus southern China, which they interpret as evidence of long-term adaptation.

A closely related study is Sexton et al. (2022), which investigates track and field athletes. They document the negative effects of high temperatures on performance and find that the sensitivity to heat varies with the temperature in the athlete's home city the days before a competition. Their focus is primarily on short-term exposure and acclimatization. Our study differs in two key ways: we examine a broader, international sample of runners rather than U.S. collegiate athletes, and we focus explicitly on long-term adaptation rather than short-term acclimatization.

The remainder of the paper is organized as follows. Section 2 provides background information, including our conceptual framework for thinking about adaptation. Section 3 describes the data and empirical strategy. Section 4 presents the empirical results. Section 5 concludes.

2 Background

This study uses data from track and field competitions organized by the World Athletics Organization. Track and field is a sport that includes running, jumping, and throwing events. We use the term "event" to refer to a specific contest, such as 100m men or 800m women. In this paper, we focus exclusively on individual running events. To better isolate individual performance, we focus exclusively on solo running events, where identifying runner performance is more straightforward than in team-based contests such as relays.

In a given running event, the races are organized into multiple rounds to determine the final winner. These rounds gradually narrow the field of competitors. The competition begins with heats: the initial races in which runners are divided into several groups based on the total number of participants. The top finishers in each heat, along with the fastest non-automatic qualifiers, advance to the next round. For events with a large number of competitors, a quarter-final round may be included. The semifinals feature the fastest competitors from the previous rounds.

The final is the championship race that determines the winners (gold, silver, and bronze medals) of the event. There are, however, some middle and long distance events (such as 3000m, 5000m, and 10,000m) which usually do not have heats, but only a final race.

The structure of these events generates an increasing degree of competitiveness, with the highest intensity occurring in the final rounds. In our empirical analysis, we distinguish between finals and preliminary rounds (heats, quarterfinals, and semifinals) to examine the potential role of incentives.

Most races in the track and field calendar occur during the spring and summer seasons. Fall and winter are generally reserved for training, with fewer competitions, typically held indoors or used as preparation for the main outdoor events. For this reason, our empirical analysis focuses on competitions held in the spring and summer.

How do we think of adaptation? We are interested in studying how adaptation shapes the impact of temperature on a runner's speed (our measure of performance). To fix ideas, consider the following model.

A runner benefits by winning a race. For simplicity, assume that the expected benefits are proportional to her speed S, which is equal to her running effort e. The benefits are then Be, where B is the marginal benefit of additional speed. A runner has a maximum speed S_i^{max}.

The cost of effort depends on the ambient temperature and the runner's sensitivity to heat. Without loss of generality, let us assume that the cost of effort is given by $C(e) = \frac{e^2}{2(1-\alpha_i t^2)}$, where t represents the deviation from an optimal temperature and α_i measures the runner's sensitivity to heat.[2] In this setting, we think of adaptation as variations in α_i, with lower values representing a runner less sensitive (i.e., better adapted) to extreme temperatures.

Using standard methods, it is straightforward to see that the dose-response, i.e. the relationship between speed and temperature, $S(t)$, would exhibit an inverted U-shape.

$$S^* = e^* == B(1 - \alpha_i t^2)$$

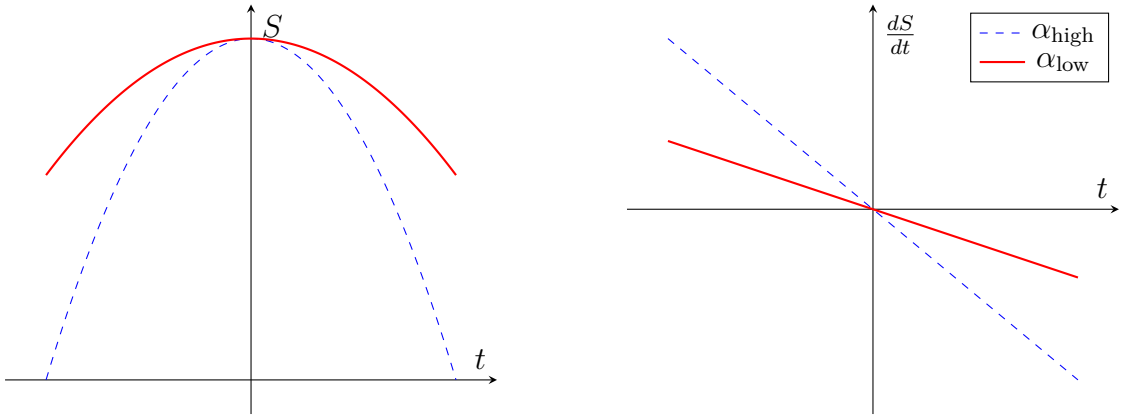
In this simplified framework, we can assess adaptation in two ways. First, we can examine the shape of the dose-response curve. Greater adaptation to heat, represented by a lower α, corresponds to a less concave curve. Second, we can analyze differences in heat sensitivity more directly by evaluating the marginal effect of temperature on speed, $\frac{dS}{dT}$. Lower sensitivity implies a flatter marginal effect curve.

Figure 1 illustrates the dose-response and marginal effect curves implied by the model

[2]We assume that $1 - \alpha_i t^2 > 0$ for all t. We choose a specific functional form for exposition purposes. Similar results can be obtained using a convex cost function that satisfies the following conditions: $\frac{dC}{de} > 0$, $\frac{d^2C}{de^2} > 0, \frac{d^2C}{dedt} > 0$ and $\frac{d^2C}{dtd\alpha_i} > 0$.

for different levels of heat sensitivity, α_i. In our empirical analysis, we estimate and display these curves.

Figure 1: Relation between speed and temperature



(a) Dose-response, $S(t)$ (b) Marginal effect, $\frac{dS}{dt}$

Notes: Figure depict dose-response $S(t) = B(1 - \alpha t^2)$ and $\frac{dS}{dt} = -2\alpha Bt$. Red solid lines represent curves for a runner with lower sensitivity (better adapted) to heat. Blue dashed lines represent curves for runner with higher sensitivty (worse adapted) to heat.

The model also suggests a potential role for incentives in shaping the dose-response relationship. When the expected benefits of winning increase, the outcome may be a corner solution in which runners aim to reach their top speed, S_i^{max}. In that case, the effect of temperature on speed diminishes. In our setting, speed becomes unresponsive to temperature because we assume a fixed maximum speed. However, this result could change if we allow the maximum speed to be affected also by temperature. In our empirical analysis, we explore this incentive mechanism by comparing the marginal effect of temperature between preliminary rounds and final races.

3 Methods

3.1 Data

Competition results Our primary data source is the online results of competitions organized worldwide by the World Athletics Organization (WAO). The data are publicly available at `https://worldathletics.org/competition/calendar-results`. We scraped the website and collected information from all available competitions up to the end of 2023.

For each race, we record the city and date, the event name (e.g., men's 100m), the round type (preliminary or final), and the results. The results include the name, finishing position, and time of each runner, along with their age, sex, and nationality. We use these data to construct a panel dataset of runners, where the unit of observation is runner i competing in race j on date t. For example, one observation corresponds to Usain Bolt (i) competing in the men's 100m final at the 2012 Summer Olympic Games in London (j) on August 5, 2012 (t).

To facilitate interpretation, we restrict our sample to individual running events, including sprints (100m, 200m, and 400m) and middle- to long-distance races (800m, 1000m, 3000m, 5000m, and 10,000m) for both men and women. We exclude indoor races, combined events (e.g., heptathlon and decathlon), road events (e.g., marathons and half marathons), relay races, blind and wheelchair races, and races with obstacles (e.g., hurdles and steeplechase). We also exclude competitions held in the fall and winter, as they are often indoor and tend to attract runners who may differ systematically from those competing in spring and summer.[3]

Measure of performance We use a runner's speed as a natural measure of performance. To account for differences across event types, we standardize speed at the event level.[4]

[3]Fall and winter competitions account for about 10% of all observations. Several elite runners often skip them to focus on training for major spring and summer events.

[4]See Table A.1 in the online appendix for average speed by event.

Specifically, we define:

$$\text{Standardized speed}_{ijt} = \frac{S_{ijt} - \max(S^{event})}{SD(S^{event})},$$

where S_{ijt} is the speed (in m/s) of runner i in race j on date t, and $\max(S^{event})$ and $SD(S^{event})$ denote the maximum and standard deviation of observed speeds within a given event, respectively.[5]

Environmental conditions We use re-analysis data from the Modern-Era Retrospective Analysis for Research and Applications version 2 (MERRA-2) project provided by NASA (GMAO, 2015). The dataset provides daily information on average weather and air pollution conditions at a spatial resolution of $0.5° \times 0.625°$. We merge the MERRA-2 data to the competition data using the geographical coordinates of the city where the competition is held and the date of each race. This procedure allows us to recover local environmental conditions on race days.

Our main measure of heat is air temperature, recorded at a height of two meters above the ground during daytime hours. We use this variable because it is widely employed in weather and climate studies.[6] However, we also test the robustness of our results using wet bulb temperature, which represents the lowest temperature air can reach through evaporative cooling. This measure provides an alternative way of assessing heat stress in humans. Additionally, we collect data on other environmental conditions, including relative humidity, precipitation rates, wind speed and direction, and fine particulate matter ($PM_{2.5}$) concentration.

Our final dataset includes over 420,000 observations covering more than 50,000 runners

[5] Our approach is similar to the performance measure used by Sexton et al. (2022), who define standardized performance as $-\frac{|T^{world} - T_{ijt}|}{SD(T^{event})}$, where T is the finishing time and T^{world} is the world record for the event. Our results remain consistent when using this alternative measure. See Figure A.6.

[6] See Figure A.1 in the online appendix for the distribution of air temperature in our sample. We restrict our regression sample to observations with daytime temperatures above 8°C.

who competed in events held across more than 1,500 cities between 2005 and 2023.[7] Table 1 presents summary statistics for the main variables used in the empirical analysis.

Table 1: Summary statistics

Variable	Mean	s.d.
A. Runner characteristics		
Speed (m/s)	7.6	1.4
Age (years)	22.5	4.6
% female	45.1	49.8
Avg. temp. in country of origin (℃)	22.4	3.5
B. Race characteristics		
% final round	63.9	48.0
% is sprint	61.0	48.8
% is home race	69.9	45.9
C.Conditions on race day		
Air temperature (℃)	21.1	5.2
Wet bulb temperature (℃)	6.6	2.2
Relative humidity (%)	32.4	8.6
Max. precipitation rate (mm/day)	28.5	72.8
East wind (m/s)	0.118	1.730
North wind (m/s)	0.063	1.622
$PM_{2.5}$ (mg/m^3)	8.9	10.6
No. obs.	420,272	

Notes: Avg. temp. country of origin refers to the average monthly temperature during summer months across all cities that hosted an athletic event in the runner's country of origin between 2014 and 2023.

[7]The number of observations increases over time, with a substantial increase in 2018. This growth reflects changes in the WAO's data collection capacity, as the organization began outsourcing this task in 2018. To address potential concerns about sampling bias, we test the robustness of our main results by excluding competitions held before 2018 (see Figure A.8).

3.2 Empirical Strategy

To study adaptation to heat, we estimate the dose-response relationship between temperature and running speed and examine how this relationship varies with previous exposure to heat. This approach aligns with the literature on adaptation and acclimatization, which assesses adaptation by evaluating how dose-response patterns change with previous exposure (Calabrese et al., 2007).

To measure previous exposure, we use the average summer temperature in a runner's country of nationality. Specifically, we collect monthly temperature data from 2014 to 2023 for all cities that hosted World Athletics competitions. For each runner, we identify their country of nationality and calculate the average city temperature during the summer months: June to September in the Northern Hemisphere and January to March in the southern hemisphere[8].

We use this variable as a proxy for long-term heat exposure during formative years, rather than short-term exposure in the days or weeks preceding a competition. This measure can capture both physiological and social dimensions of adaptation to heat. However, two key limitations may introduce measurement error. First, a runner's nationality may not reflect their actual place of residence or training, and thus may not represent recent heat exposure. Second, some athletes change their nationality to compete for another country,[9] meaning that nationality may not correspond to the climatic conditions of their birthplace. Due to data limitations, we cannot test the robustness of our results using alternative measures based on the place of residence or training.

[8]Our measure of previous exposure is comparable to recent research on heat effects in athletics, such as Burke et al. (2023), which uses the temperature in the birthplaces of tennis players. However, it differs from studies such as Qiu and Zhao (2022) and Sexton et al. (2022), which instead use weather conditions at the athlete's place of residence or training.

[9]For example, Bernard Lagat, born in Kenya, later became a U.S. citizen.

Baseline regression To estimate the dose-response function, we use the following regression model:

$$\text{standardized speed}_{ijt} = f(\beta, W_{jt}, C_i) + \gamma X_{ijt} + \eta_i + \epsilon_{ijt}, \tag{1}$$

where the unit of observation is runner i in race j on date t. W_{jt} is a set of local environmental conditions that includes temperature, relative humidity, precipitation rate, wind speed, and daily levels of $PM_{2.5}$. C_i represents the average summer temperature in the runner's country of nationality, our measure of a runner's previous exposure to heat. η_i are a runners' fixed effect and X_{jt} includes other runner and race characteristics such as indicators for whether the race is a home event or a final, the duration of the competition, and event-year fixed effects.

Our specification identifies the relationship between temperature and speed by exploiting within-runner variation. We also include a rich set of time-varying controls and fixed effects, allowing flexible time trends to vary by event. These controls help mitigate potential biases due to omitted variables.

To examine adaptation, we include an interaction term between temperature (and its square) and C_i. Our specification allows for non-linearities and complementarities between local environmental conditions, as previous research has found these factors to be relevant (Li et al., 2015; Sillmann et al., 2021). In particular, the function f is a second-degree polynomial that includes quadratic terms and cross-terms that capture interactions among environmental conditions.

4 Results

4.1 Main results

Figure 2 presents our main results. To facilitate exposition, we display the predicted dose-response in panel (a) and the marginal effect, i.e., the slope of the dose-response, in panel (b).[10] These figures are the empirical correlates of the functions derived in Figure 1. The top row displays the predicted dose-response for the average runner. The bottom row illustrates the heterogeneity in the dose-response with respect to the previous heat exposure of the runners. Specifically, it shows the predicted dose-response assuming C_i is set to the 10th and 90th percentiles of the countries' summer temperatures, corresponding to cold and hot countries, respectively.

We observe an inverted U-shaped relationship between speed and temperature. Speed increases with temperature, reaches a peak at around 30 ℃ and then declines at higher temperatures. The same negative impact of extreme heat can be observed in Panel (b), which shows a diminishing marginal effect that starts positive but eventually becomes negative. These results are consistent with previous findings and, while not surprising, increase confidence that our empirical approach captures a meaningful relationship between heat and performance.

Our main finding emerges when we examine the heterogeneity in responses to heat based on previous exposure (Figure 1, bottom row). Among runners from warmer countries, the dose-response relationship is considerably flatter, and the marginal effect of higher temperatures is positive or close to zero. In contrast, for runners from colder countries, exposure to extreme heat has a significantly detrimental effect. We interpret these differences in heat sensitivity as evidence of adaptation to climatic conditions.

[10] We use the Stata command `margins` to calculate predicted values and marginal effects (dydx), setting temperature values from 8 to 40 ℃ while holding other variables at their mean values. Table A.2 in the appendix presents all estimates of the baseline regression.

Adaptation appears to play a quantitatively important role. Our estimates imply that increasing race-day temperature from 30 ℃ to 40 ℃ would reduce running speed by nearly 0.10 standard deviations for runners from cold countries, while having almost no effect on runners from hot countries. This performance difference is comparable in magnitude to the typical gap between gold and silver medalists.

These results are robust to a range of specification changes, including controlling for the altitude of the competition venue, using wet bulb temperature instead of air temperature, and replacing summer temperature with annual temperature (see Figures A.3 to A.6 in the appendix). However, when we use a measure of short-term exposure similar to Sexton et al. (2022), we find no evidence of differences in heat sensitivity between runners (see Figure A.6 in the appendix).[11]

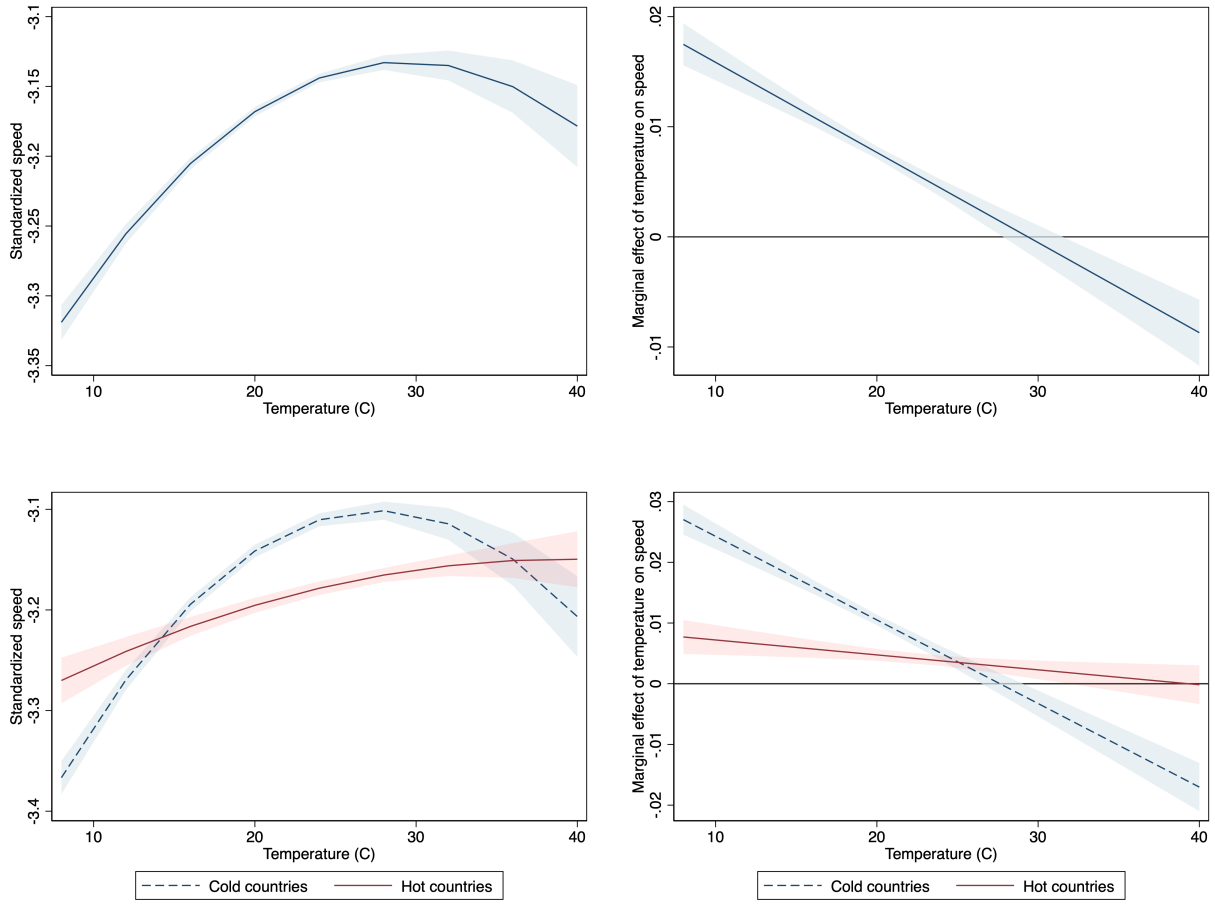
4.2 Adaptation by type of race

We next examine how heat stress and adaptation affect performance across different race types. First, we compare heat sensitivity in sprint events (100m to 400m) with that in medium- to long-distance races (800m to 15,000m) (see Figure 3). We focus on race distance because prior research suggests that aerobic activities, such as endurance races, are more sensitive to heat stress (Otani et al., 2018; Bouscaren et al., 2019).

Consistent with these findings, we observe that performance in longer races begins to decline at significantly lower temperatures than in sprints (23℃ compared to 30 ℃). Moreover, the gap in heat sensitivity between runners from hot and cold countries is more pronounced in endurance events. In those races, athletes from cold climates experience substantial performance declines under heat stress, whereas runners from hot climates appear largely unaffected. We interpret this pattern as suggestive evidence that adaptation plays a more

[11] We replace the average summer temperature in the country of nationality with the temperature recorded there 7 to 14 days before a competition.

Figure 2: Temperature and runner's speed



(a) Dose-response

(b) Marginal effect

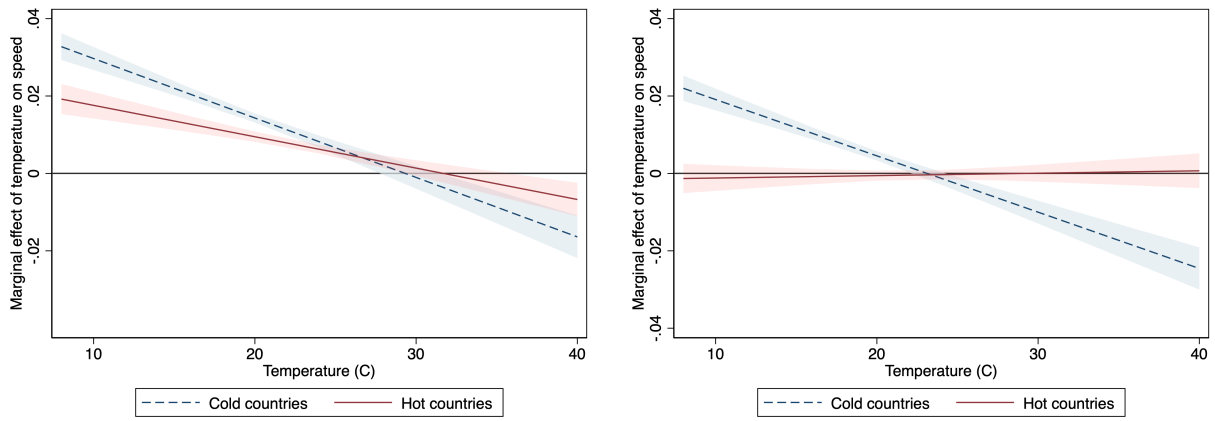
Notes: Figures display the predicted relationship between temperature and standardized speed computed using the regression estimates. Shaded areas represent the 95% confidence interval. Panel (a) presents the predicted dose-response curve, while panel (b) shows the marginal effect of additional temperature, i.e., the slope of panel (a). The top row predicts the relationship for the average country, whereas the bottom row considers two scenarios assuming different summer temperatures of the runner's country of origin. The "cold country" scenario uses the 10th percentile of temperatures (18.2 ℃), while the "hot country" scenario uses the 90th percentile (26.6 ℃).

important role in aerobic activities.

Second, we compare heat sensitivity between preliminary rounds (heats, quarterfinals and semifinals) and final races (see Figure 4). Our main observation is that, for all runners, performance is more sensitive to heat in the preliminary rounds than in the finals.

One interpretation of this pattern is that runners may adjust their effort to manage heat stress. They conserve energy and avoid injury during preliminary rounds while reserving maximum effort for the finals.[12] As noted above, athletes in preliminary rounds only need to run fast enough to qualify. In contrast, final races carry higher stakes, as they determine medal outcomes. If this interpretation holds, it suggests that the observed sensitivity to heat reflects not only physiological limits but also a strategic response to incentives.

Figure 3: Marginal effect of temperature, by race distance



(a) Sprints

(b) Medium and long-distance

Notes: Figures display the predicted marginal effect of additional temperature on standardized speed across different race types. Sprints include races up to 400 meters, while middle- and long-distance races cover distances from 800 to 15,000 meters.

4.3 Alternative explanations

In this section, we examine two alternative explanations for our findings: confounding factors and composition effects. A first possible explanation is that our results reflect heterogeneous dose-responses driven by factors correlated with the country of nationality of a runner, rather than genuine adaptation to heat. For example, runners from hotter countries may differ in

[12]There is evidence that individuals can reduce the adverse effects of heat stress by increasing psychological effort (Barwood et al., 2008).

access to healthcare systems, training facilities, or economic incentives. These factors could influence their sensitivity to heat independently of their country's average temperature. If this is the case, interpreting the results as evidence of adaptation would be misleading.

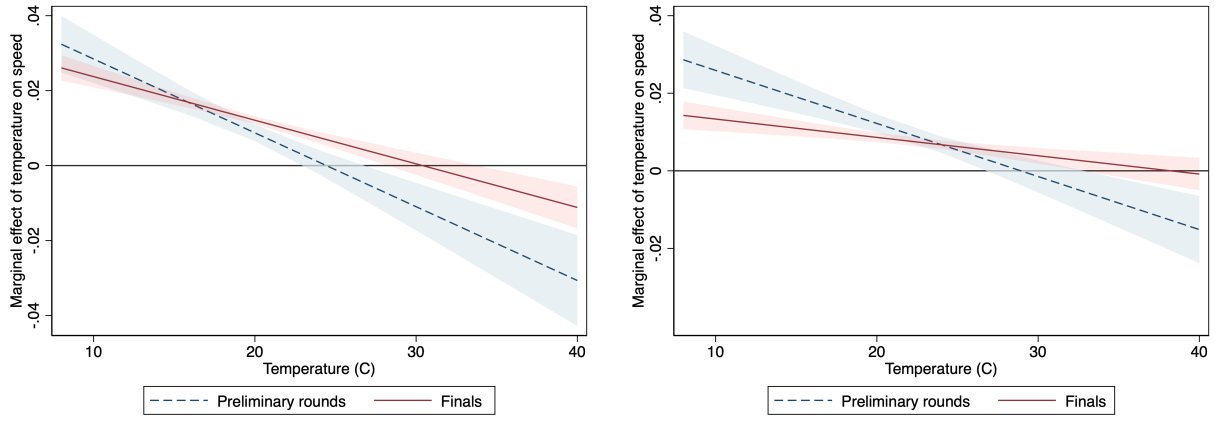
We partially address this concern by interacting temperature with a country's per capita GDP, which serves as a rough proxy for these potential confounding factors. Although this variable does not capture the full range of relevant heterogeneity, it is reassuring that our main results remain robust. In particular, we continue to observe consistent differences in heat sensitivity between runners from hot and cold countries (see Figure A.2 in the appendix).

A second possible explanation is that the results are driven by composition effects. Runners from hot countries may systematically differ from those from cold countries. For instance, only the most resilient or highest-performing athletes from hotter regions may qualify for competition, introducing selection bias. Composition could also shift due to systematic attrition. For example, on hotter days, top runners from cold countries might be more likely to withdraw from competition. In both cases, we would observe differences in the average heat sensitivity between groups that are not related to prior exposure or adaptation.

To assess this possibility, we replicate our baseline analysis in subsamples defined by runners' characteristics. This approach allows us to compare dose-response patterns while holding certain observable traits constant. Specifically, we focus on two key attributes: sex and elite status. To measure elite status, we calculate each runner's average speed in a given event (across all observed competition). Then we rank runners by their average speed, and divide the sample into the top 20 percent and the remaining 80 percent.

The results, presented in Figure 5, closely resemble our baseline findings. We continue to observe sizable differences in heat sensitivity between runners from hot and cold countries. This pattern reduces concerns that compositional differences alone account for our results, although we acknowledge that we cannot fully rule out selection on unobservable characteristics.

Figure 4: Marginal effect of temperature, preliminary rounds vs finals

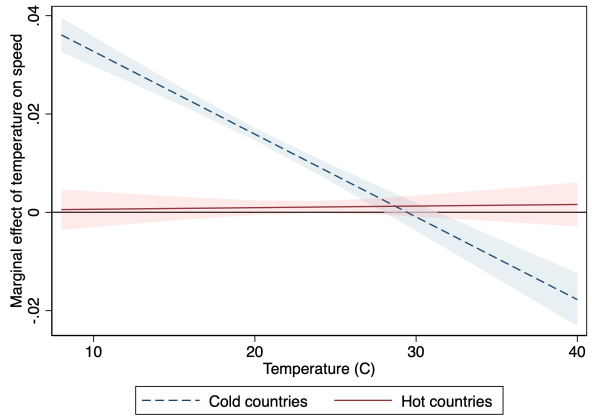


(a) Cold countries

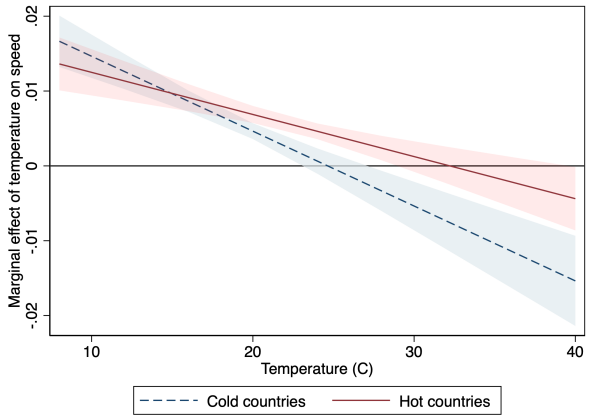
(b) Hot countries

Notes: Figures display the predicted marginal effect of additional temperature on standardized speed for preliminary rounds (heats, quarterfinals and semifinals) and final races. Figure exclude races of 3,000 meters or longer, as such races typically do not have preliminary rounds. Panels (a) and (b) assume different summer temperatures of the runner's country of origin (cold and hot countries) as in Figure 2.

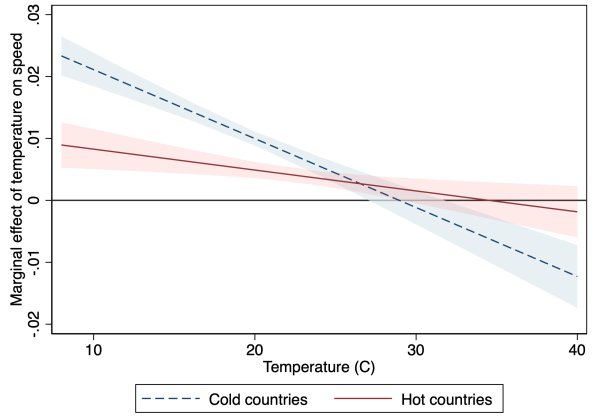
Figure 5: Marginal effect of temperature, by type of runner



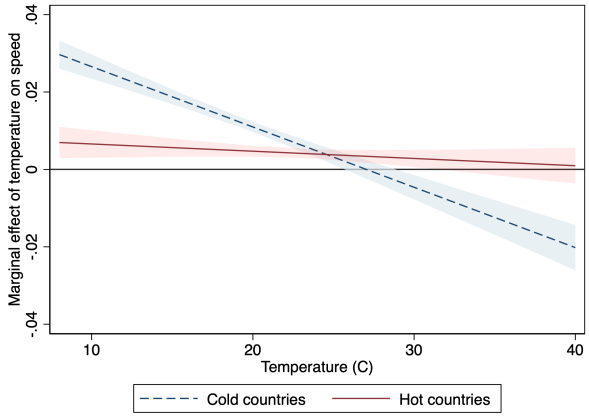
(a) Top 20% runners



(b) Bottom 80% runners



(c) Female



(d) Male

Notes: Figures show the predicted marginal effect of additional temperature on standardized speed across different types of runners. Panels (a) and (b) classify runners based on the distribution of their average speed within an event (i.e. men 100m), with Panel (a) corresponding to the faster runners. Panels (c) and (d) separate runners by sex.

5 Conclusion

This paper provides new evidence on how long-term adaptation to heat influences human performance at extreme temperatures. Using a global dataset of competitive runners, we show that the adverse effects of high temperatures on performance vary systematically with the climate of a runner's country of origin. In particular, runners from warmer countries are significantly less affected by extreme heat, especially in endurance events.

These findings offer two key insights. First, they provide large-scale, real-world evidence that long-term exposure can build resilience to heat, reinforcing the notion that human performance is shaped not only by biological limits but also by environmental experience. Second, the observed variation in heat sensitivity, modulated by both climate and incentives, highlights the potential productivity losses from extreme heat and the scope for mitigating them through adaptation.

The broader economic implication is that climate change may affect labor productivity unevenly, with higher costs in settings where workers lack prior exposure or adaptation to high temperatures. As rising global temperatures increase the frequency of heat extremes, these productivity losses could impose substantial economic burdens, particularly in sectors reliant on sustained physical effort.

References

Barreca, Alan, Karen Clay, Olivier Deschenes, Michael Greenstone, and Joseph S Shapiro, "Adapting to climate change: The remarkable decline in the US temperature-mortality relationship over the twentieth century," *Journal of Political Economy*, 2016, *124* (1), 105–159.

Barwood, Martin, Richard Thelwell, and Michael Tipton, "Psychological skills training improves exercise performance in the heat," *Medicine+ Science in Sports+ Exercise*, 2008, *40* (2), 387.

Blanchfield, Anthony W, James Hardy, Helma M De Morree, Walter Staiano, Samuele Maria Marcora et al., "Talking yourself out of exhaustion: the effects of self-talk on endurance performance," *Med Sci Sports Exerc*, 2014, *46* (5), 998–1007.

Bouscaren, Nicolas, Guillaume Y Millet, and Sebastien Racinais, "Heat stress challenges in marathon vs. ultra-endurance running," *Frontiers in Sports and Active Living*, 2019, *1*, 59.

Burke, Marshall, Vincent Tanutama, Sam Heft-Neal, Miyuki Hino, and David Lobell, "Game, sweat, match: Temperature and elite worker productivity," Technical Report, National Bureau of Economic Research 2023.

Calabrese, Edward J, Kenneth A Bachmann, A John Bailer, P Michael Bolger, Jonathan Borak, Lu Cai, Nina Cedergreen, M George Cherian, Chuang C Chiueh, Thomas W Clarkson et al., "Biological stress response terminology: integrating the concepts of adaptive response and preconditioning stress within a hormetic dose–response framework," *Toxicology and applied pharmacology*, 2007, *222* (1), 122–128.

Carleton, Tamma, Amir Jina, Michael Delgado, Michael Greenstone, Trevor Houser, Solomon Hsiang, Andrew Hultgren, Robert E Kopp, Kelly E McCusker, Ishan Nath et al., "Valuing the global mortality consequences of climate change accounting for adaptation costs and benefits," *The Quarterly Journal of Economics*, 2022, *137* (4), 2037–2105.

Evans, Gary W, Stephen V Jacobs, and Neal B Frager, "Human adaptation to smog," *Journal of the Air Pollution Control Association*, 1982, *32* (10), 1054–1057.

Gibson, Oliver R, Carl A James, Jessica A Mee, Ashley GB Willmott, Gareth Turner, Mark Hayes, and Neil S Maxwell, "Heat alleviation strategies for athletic performance: a review and practitioner guidelines," *Temperature*, 2020, *7* (1), 3–36.

GMAO, "MERRA-2 inst1_2d_asm_Nx: 2d, 1-Hourly, Instantaneous, Single-Level, Assimilation, Single-Level Diagnostics V5.12.4," 2015. Global Modeling and Assimilation Office (GMAO). Goddard Earth Sciences Data and Information Services Center (GES DISC). Accessed November 2024.

Li, Li, Jun Yang, Cui Guo, Ping-Yan Chen, Chun-Quan Ou, and Yuming Guo, "Particulate matter modifies the magnitude and time course of the non-linear temperature-mortality association," *Environmental Pollution*, 2015, *196*, 423–430.

Otani, Hidenori, Mitsuharu Kaya, Akira Tamaki, Heita Goto, Takayuki Goto, and Minayuki Shirato, "Diurnal effects of prior heat stress exposure on sprint and endurance exercise capacity in the heat," *Chronobiology International*, 2018, *35* (7), 982–995.

Qiu, Yun and Jinhua Zhao, "Adaptation and the distributional effects of heat: Evidence from professional archery competitions," *Southern Economic Journal*, 2022, *88* (3), 1149–1177.

Schlenker, Wolfram and Michael J Roberts, "Nonlinear temperature effects indicate severe damages to US crop yields under climate change," *Proceedings of the National Academy of sciences*, 2009, *106* (37), 15594–15598.

Sexton, Steven, Zhenxuan Wang, and Jamie T Mullins, "Heat adaptation and human performance in a warming climate," *Journal of the Association of Environmental and Resource Economists*, 2022, *9* (1), 141–163.

Sillmann, Jana, Kristin Aunan, Lisa Emberson, Patrick Büker, Bob Van Oort, Connie O'Neill, Noelia Otero, Divya Pandey, and Anouk Brisebois, "Combined impacts of climate and air pollution on human health and agricultural productivity," *Environmental Research Letters*, 2021, *16* (9), 093004.

Zhao, Mengzhen, Jason Kai Wei Lee, Tord Kjellstrom, and Wenjia Cai, "Assessment of the economic impact of heat-related labor productivity loss: a systematic review," *Climatic Change*, 2021, *167* (1), 22.

ONLINE APPENDIX

A Additional Figures and Tables

Table A.1: Average speed (m/s) by race event

Distance (m)	Women	Men
100	8.264	9.267
200	8.074	9.170
400	7.407	8.605
800	7.175	8.284
1,500	6.127	7.117
3,000	5.584	6.455
5,000	5.143	5.958
10,000	5.040	5.820

Notes: Average speed for all races in dataset.

Table A.2: Regression results

	Dep. Variable = standardized speed				
		Type of race			
	All sample	Sprints	Medium & long distance	Preliminary rounds	Finals
	(1)	(2)	(3)	(4)	(5)
Temperature	0.096***	0.086***	0.101***	0.131***	0.072***
	(-0.007)	(-0.01)	(-0.01)	(-0.014)	(-0.010)
Temperature^2	-0.002***	-0.002***	-0.002***	-0.002***	-0.001***
	(-0.0001)	(-0.0002)	(-0.0002)	(-0.0003)	(-0.0002)
Temp. × country's summer temp.	-0.00338***	-0.00230***	-0.00422***	-0.00446***	-0.00213***
	(-0.0003)	(-0.0005)	(-0.0004)	(-0.0006)	(-0.0004)
Temp.^{2} × country's summer temp.	0.00007***	0.00004***	0.00009***	0.00007***	0.00004***
	(-0.00001)	(-0.00001)	(-0.00001)	(-0.00001)	(-0.00001)
Country's summer temp.	0.0342***	0.0219***	0.0434***	0.0577***	0.0191***
	(-0.004)	(-0.005)	(-0.005)	(-0.008)	(-0.005)
R-squared	0.813	0.793	0.827	0.814	0.827
No. runners	52,427	30,245	23,455	18,019	22,547
No. observations	420,272	255,191	162,401	141,054	212,894

Notes: Standard errors (in parenthesis) are clustered at the runner level. * denotes significant at 10%, ** significant at 5% and *** significant at 1%. All regressions include runner and year-by-event fixed effects plus length of competition (no. days), and indicators of being a home competition, and a final race. Additional environmental controls include: average daytime levels of pollution ($PM_{2.5}$), humidity, north and east winds' speeds, and precipitation, as well as their quadratic terms and interaction between temperature with pollution, humidity, and precipitation, and pollution with wind speeds and precipitation. Columns 2 to 7 split the sample by type of race or runner. Columns 8 and 9 add interaction terms of temperature with the average GPD per capita in the runner's country of nationality. Temperature = average daytime temperature on race day. Country's summer temp.= average daytime summer temperature in the runner's country of nationality.

Table A.2: Regression results (cont.)

	Dep. Variable = standardized speed			
	Elite status		Additional confounders	
	Top 20% runners	Bottom 80% runners	Country's GDP	City elevation
	(6)	(7)	(8)	(9)
Temperature	0.150***	0.039***	0.113***	0.093***
	(-0.011)	(-0.011)	(-0.021)	(-0.008)
Temperature^2	-0.003***	-0.001***	-0.003***	-0.002***
	(-0.0002)	(-0.0003)	(-0.0005)	(-0.0002)
Temp. × country's summer temp.	-0.00587***	-0.0008	-0.0034***	-0.0033***
	(-0.0005)	(-0.0005)	(-0.0004)	(-0.0003)
Temp.^{2} × country's summer temp.	0.00010***	0.00003*	0.00007***	0.00006***
	(-0.00001)	(-0.00001)	(-0.00001)	(-0.00001)
Country's summer temp.	0.0716***	-0.003	0.0338***	0.0346***
	(-0.006)	(-0.005)	(-0.004)	(-0.004)
R-squared	0.640	0.757	0.814	0.813
No. runners	15,354	40,520	47,355	30,245
No. observations	201 091	216 504	379 641	420 272

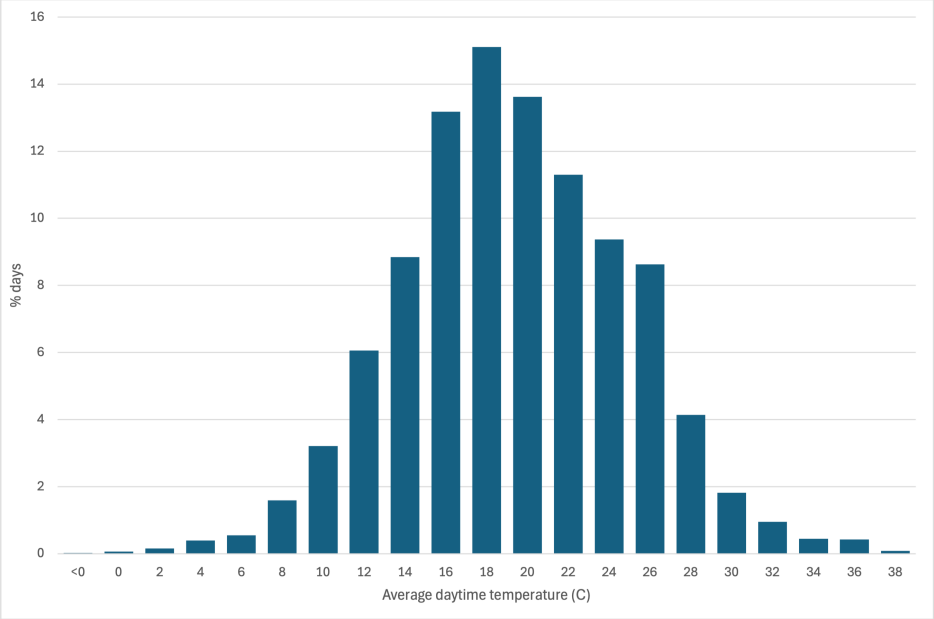
Notes: Standard errors (in parenthesis) are clustered at the runner level. * denotes significant at 10%, ** significant at 5% and *** significant at 1%. All regressions include runner and year-by-event fixed effects plus length of competition (no. days), and indicators of being a home competition, and a final race. Additional environmental controls include: average daytime levels of pollution ($PM_{2.5}$), humidity, north and east winds' speeds, and precipitation, as well as their quadratic terms and interaction between temperature with pollution, humidity, and precipitation, and pollution with wind speeds and precipitation. Columns 2 to 7 split the sample by type of race or runner. Columns 8 and 9 add interaction terms of temperature with the average GPD per capita in the runner's country of nationality. Temperature = average daytime temperature on race day. Country's summer temp.= average daytime summer temperature in the runner's country of nationality.

Table A.3: Alternative measures of exposure

	Dep. Variable = standardized speed	
	Using annual country temp. (1)	Sexton et al. (2022) measure (2)
Temperature	0.048***	0.027***
	(-0.004)	(-0.004)
Temperature^2	-0.001***	-0.0005***
	(-0.000)	(-0.0001)
Temp. × country's annual temp.	-0.002***	
	(-0.0003)	
Country annual temp	0.015***	
	(-0.003)	
Temp.^{2} × country's annual temp.	0.00004***	
	(-0.0000)	
Temp. × country's temp. previous 7-14 days		-0.0006**
		(-0.0002)
Country's temp. previous 7-14 days		0.013***
		(-0.002)
Temp.^{2} × country's temp. previous 7-14 days		0.00001
		(-0.000)
R-squared	0.813	0.810
No. runners	52,460	47,237
No. observations	420,559	378,484

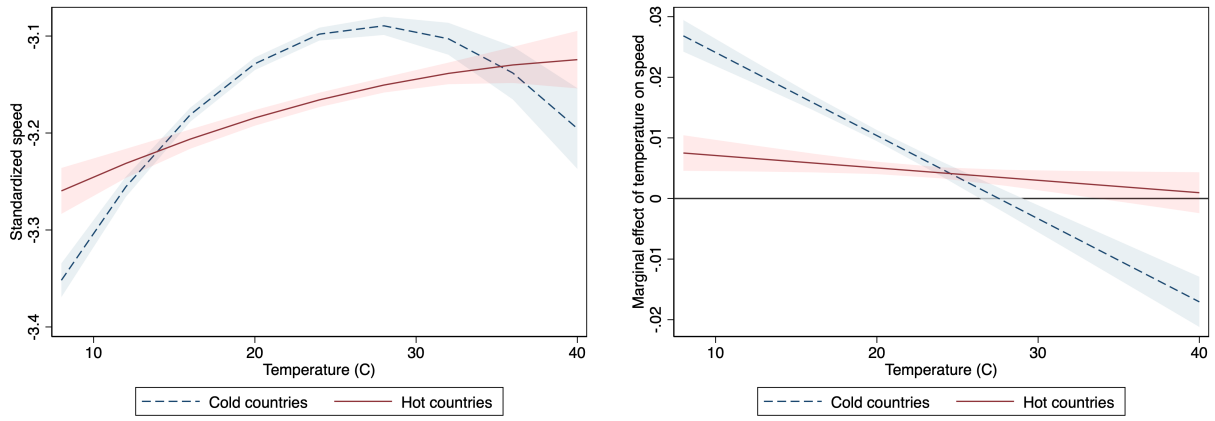
Notes: Standard errors (in parenthesis) are clustered at the runner level. * denotes significant at 10%, ** significant at 5% and *** significant at 1%. All regressions include runner and year-by-event fixed effects plus length of competition (no. days), and indicators of being a home competition, and a final race. Additional environmental controls include: average daytime levels of pollution ($PM_{2.5}$), humidity, north and east winds' speeds, and precipitation, as well as their quadratic terms and interaction between temperature with pollution, humidity, and precipitation, and pollution with wind speeds and precipitation. Column 1 replaces the mean summer temperature in the runner's country of nationality with the mean annual temperature. Column 2 follows an approach similar to Sexton et al. (2022). It uses the mean temperature in the runner's country of nationality in the days 7 to 14 before the competition.

Figure A.1: Distribution of average daytime temperature (℃)



Notes: Figure shows the proportion of day-city observations in various 2-℃ degree bins.

Figure A.2: Robustness checks - adding per capita GDP

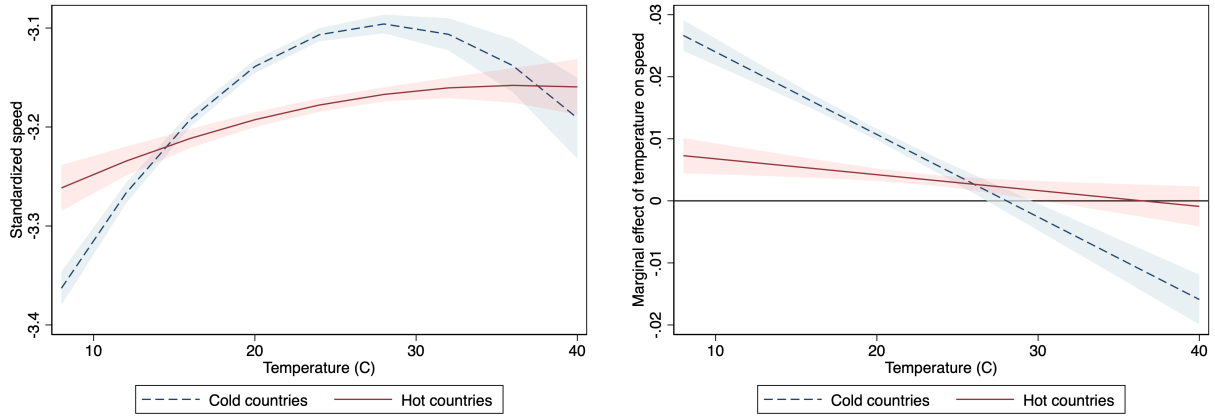


(a) Dose-response

(b) Marginal effect

Notes: Figures show the predicted dose response (panel a) and the marginal effect of additional temperature on standardized speed (panel b). The regression model uses the baseline specification but includes the interaction of temperature (and its square) and the logarithm of the country of origin's per capita GDP, calculated as the average over the period 2019-2023

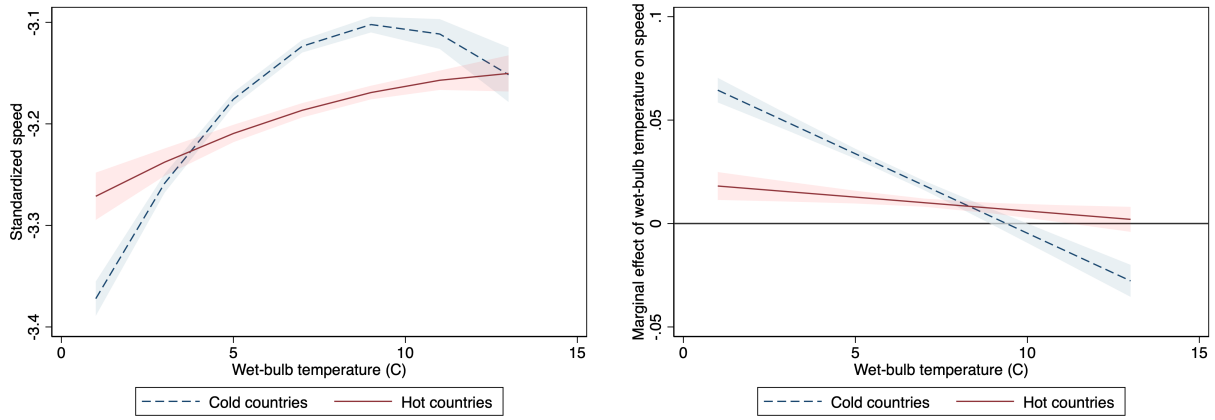
Figure A.3: Robustness checks - adding altitude



(a) Dose-response (b) Marginal effect

Notes: Figures show the predicted dose response (panel a) and the marginal effect of additional temperature on standardized speed (panel b). The regression model uses the baseline specification but includes the interaction of temperature (and its square) and altitude (in meters above sea level) of the city where the competition took place.

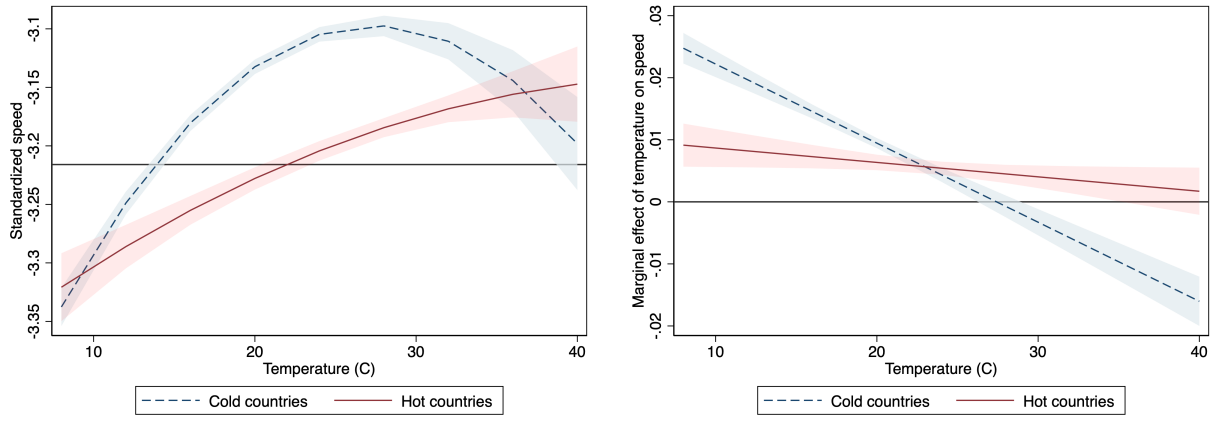
Figure A.4: Robustness checks - using wet bulb temperature



(a) Dose-response (b) Marginal effect

Notes: Figures show the predicted dose response (panel a) and the marginal effect of additional wet bulb temperature on standardized speed (panel b). The regression model uses the baseline specification but replaces the air tempetature in the competition day for wet bulb temperature.

Figure A.5: Robustness checks - using annual country's temperature

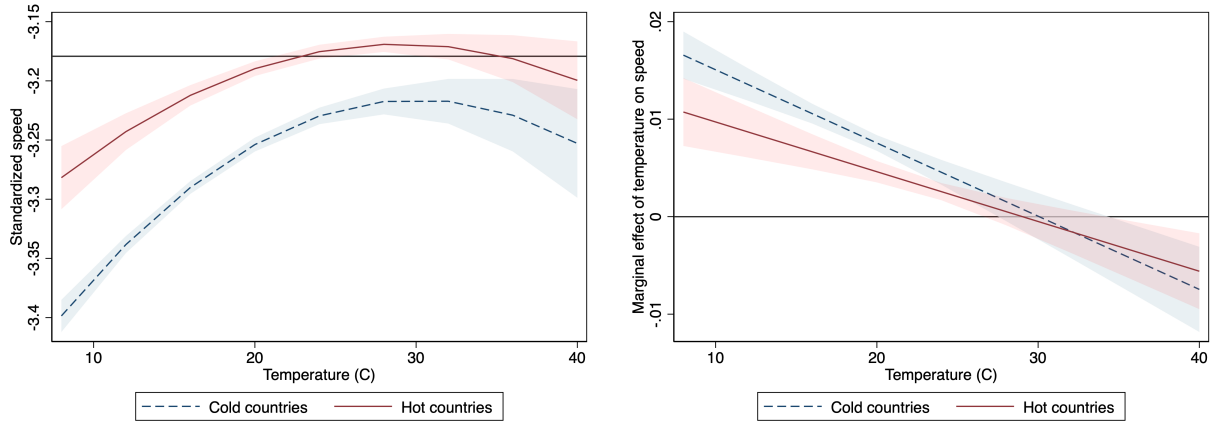


(a) Dose-response

(b) Marginal effect

Notes: Figures show the predicted dose response (panel a) and the marginal effect of additional temperature on standardized speed (panel b). The regression model uses the baseline specification, but replaces the average summer temperature of the country of origin for the average annual temperature in that country. The "cold country" scenario uses the 10th percentile of temperatures (8.7 ℃), while the "hot country" scenario uses the 90th percentile (20.4 ℃).

Figure A.6: Robustness checks - using recent country's temperature

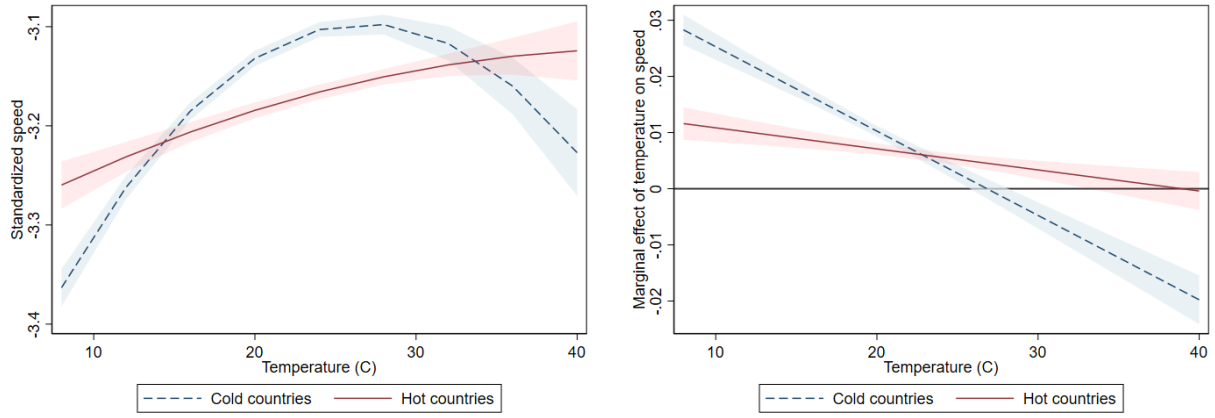


(a) Dose-response

(b) Marginal effect

Notes: Figures show the predicted dose response (panel a) and the marginal effect of additional temperature on standardized speed (panel b). The regression model uses the baseline specification, but replaces the average summer temperature of the country of origin for the average temperature in that country in days 7-13 before the competition. The "cold country" scenario uses the 10th percentile of temperatures (13.8 ℃), while the "hot country" scenario uses the 90th percentile (26.7 ℃).

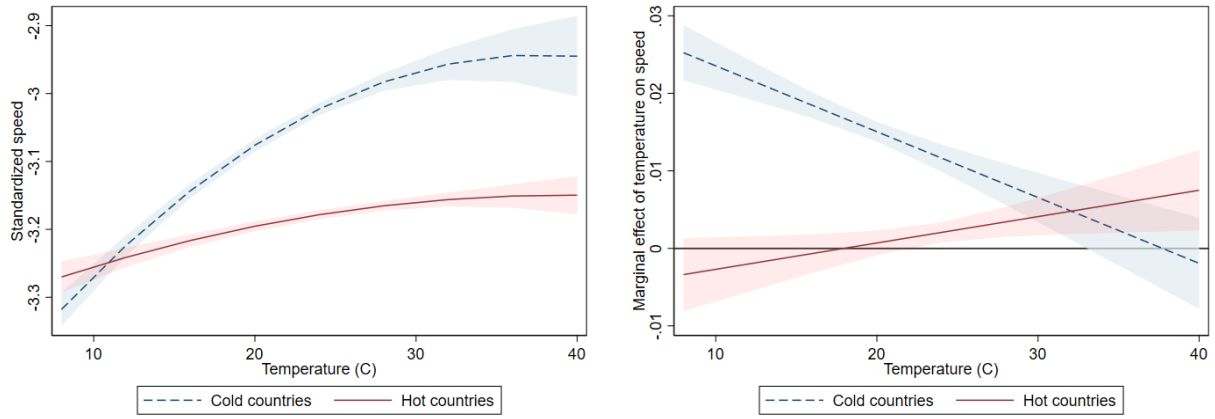
Figure A.7: Robustness checks - CV controls



(a) Dose-response (b) Marginal effect

Notes: Figures show the predicted dose response (panel a) and the marginal effect of additional temperature on standardized speed (panel b). The regression model uses the baseline specification, but includes the weather disparity in the country of origin . The "cold country" scenario uses the 10th percentile of temperatures (13.8 ℃), while the "hot country" scenario uses the 90th percentile (26.7 ℃).

Figure A.8: Robustness checks - exclusing competitions before 2018



(a) Dose-response (b) Marginal effect

Notes: Figures show the predicted dose response (panel a) and the marginal effect of additional temperature on standardized speed (panel b). The regression model uses the baseline specification, but excludes the competitions taken place before 2018. The "cold country" scenario uses the 10th percentile of temperatures (13.8 ℃), while the "hot country" scenario uses the 90th percentile (26.7 ℃).